

COTOBATO, Philippines

WMO: 987460

Lat: 7.162N

Lon: 124.215E

Elev: 203

StdP: 14.59

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	74.3	75.0	68.6	106.0	80.3	69.7	109.9	78.6	12.1	89.4	11.3	88.9	3.2	180	0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	14.6	93.8	80.4	92.8	80.4	91.8	80.4	82.3	91.7	81.8	91.1	81.3	90.5	6.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	152.4	90.4	78.8	150.6	90.1	78.3	148.1	89.4	46.2	92.3	45.7	91.4	45.2	90.8	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.8	9.5	DB	72.2	95.9	3.2	2.5	69.9	97.7	68.1	99.2	66.3	100.6	63.9	102.4	(4)
(5)			WB	69.4	83.7	1.7	1.6	68.2	84.9	67.3	85.8	66.3	86.7	65.1	87.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.2	82.7	83.5	84.2	84.5	83.9	82.9	82.3	82.6	82.4	82.9	83.1	83.3
	DBStd	1.76	1.75	1.52	1.66	1.57	1.79	1.67	1.63	1.54	1.63	1.60	1.48	1.56
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12111	1013	938	1060	1034	1050	987	1002	1010	972	1018	994	1033
	CDD65	6636	548	518	595	584	585	537	537	545	522	553	544	568
	CDH74	73852	5882	5801	6930	6988	6769	5911	5690	5891	5582	6026	6049	6333
	CDH80	27963	2056	2212	2859	3018	2769	2173	1950	2096	1940	2213	2277	2400
Wind	WSAvg	4.6	4.7	5.1	4.9	4.9	4.7	4.5	4.4	4.5	4.4	4.5	4.6	4.6
Precipitation	PrecAvg	91.6	3.6	3.9	4.3	5.3	9.6	11.4	10.6	8.5	9.9	10.6	7.0	5.2
	PrecMax	138.2	7.9	10.4	11.5	15.7	21.3	24.4	27.0	14.3	23.9	24.2	14.2	16.7
	PrecMin	47.5	0.0	0.0	0.0	1.7	2.3	5.6	0.0	3.7	2.9	3.0	1.5	1.3
	PrecStd	27.2	2.0	2.6	3.5	3.7	5.2	4.8	6.2	2.6	5.6	5.4	3.2	3.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.8	93.7	94.8	95.3	95.7	93.1	92.4	92.6	92.6	93.3	93.0	93.5
		MCWB	80.3	79.5	79.5	80.1	81.0	80.7	81.2	81.1	81.1	80.1	81.2	80.4
	2%	DB	91.1	91.9	92.9	93.3	93.2	91.3	90.7	90.9	90.9	91.4	91.4	91.7
		MCWB	79.9	79.5	79.7	80.6	80.9	80.6	80.3	80.5	80.5	80.4	80.7	80.4
	5%	DB	89.7	90.6	91.6	91.8	91.4	89.9	89.3	89.6	89.6	90.0	90.1	90.4
		MCWB	79.5	79.5	79.7	80.6	80.7	80.2	79.8	80.0	79.9	80.1	80.3	80.1
	10%	DB	88.0	89.1	90.1	90.5	89.8	88.6	87.9	88.3	88.0	88.6	88.8	89.1
		MCWB	79.2	79.2	79.6	80.4	80.3	79.8	79.3	79.5	79.4	79.7	79.9	79.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	81.9	82.4	82.7	82.5	81.9	81.9	82.2	82.1	82.0	82.2	82.0
		MCDB	90.9	91.3	91.9	92.5	92.0	90.9	90.9	91.3	91.3	91.1	91.6	91.3
	2%	WB	80.9	81.0	81.4	81.9	81.7	81.1	80.9	81.1	81.0	81.0	81.3	81.2
		MCDB	89.5	90.1	90.9	91.4	91.4	89.9	89.5	89.8	89.7	89.9	90.4	90.4
	5%	WB	80.0	80.2	80.7	81.2	81.0	80.5	80.1	80.4	80.2	80.4	80.7	80.6
		MCDB	88.4	89.1	90.0	90.6	90.3	89.1	88.4	88.9	88.5	89.0	89.4	89.5
	10%	WB	79.2	79.4	79.9	80.5	80.3	79.8	79.3	79.6	79.4	79.7	79.9	79.8
		MCDB	87.3	88.1	88.9	89.7	89.3	87.9	87.3	87.7	87.4	88.0	88.2	88.4
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.4	14.1	14.6	14.1	13.1	12.5	13.0	12.7	13.2	13.6	13.7
		MCDBR	14.6	15.0	15.5	15.7	15.8	14.6	14.3	14.6	14.7	14.7	14.8	15.1
		MCWBR	7.3	7.2	7.1	7.5	7.4	7.6	7.3	7.6	7.5	7.3	7.6	7.3
	5% WB	MCDBR	13.7	13.9	14.3	15.0	14.9	13.9	13.7	14.0	13.9	14.0	14.3	14.5
		MCWBR	7.7	7.7	7.8	8.1	7.8	7.6	7.5	7.7	7.7	7.6	7.7	7.7
Clear-Sky Solar Irradiance	taub	0.395	0.400	0.411	0.431	0.422	0.406	0.406	0.427	0.431	0.452	0.417	0.404	
	taud	2.522	2.489	2.464	2.402	2.448	2.500	2.498	2.421	2.398	2.341	2.456	2.492	
	Ebn at Noon	292	293	289	279	276	277	278	276	279	274	283	287	
	Edh at Noon	34	36	37	39	36	34	34	38	39	41	36	34	
All-Sky Solar Radiation	RadAvg	1473	1674	1806	1842	1679	1583	1545	1590	1599	1586	1557	1505	
	RadStd	128	137	126	88	92	70	97	99	122	112	112	100	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	+0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page